Appendix R3: Basic Models

WLS, GLM, Logistic and Trees

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This Quarto document contains R code examples for basic predictive models, including Ordinary Least Squares (OLS), Weighted Least Squares (WLS), Generalized Linear Models (GLM), Binary Logistic regression, and regression and classification trees. For the corresponding conceptual material associated with these appendinces, consult the main book Predictive Analytics and Machine Learning for Managers. Chat with the PAML4M book and its related lecture slides and R scripts with its companion PAML4M ChatBot.

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1. OLS Assumptions

Residuals are Normally Distributed

QQ Plot

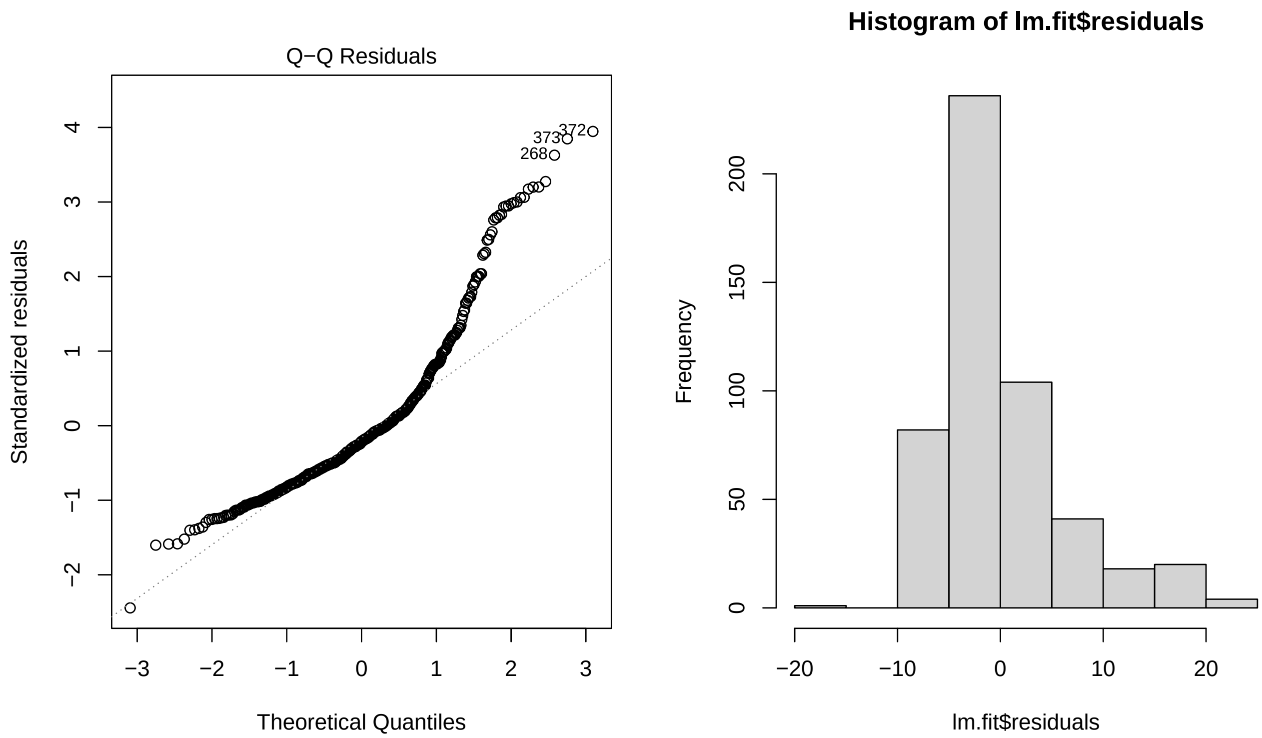
When we fit an OLS model, we need to inspect the residuals for normality. Let's start by fitting an OLS linear model

```
library(MASS) # To read the Boston housing market data
options(scipen = 4) # To minimize the use of scientific notation
lm.fit <-lm(medv ~ lstat,
```

```
              data = Boston) # Fit the model and store results in lm.ols
```

Now, let's visually inspect the normality of the residuals.

```
par(mfrow = c(1, 2))
plot(lm.fit, which = 2)
hist(lm.fit$residuals)
```




```
par(mfrow = c(1, 1))
```

We can see in the plot and histogram above, that the residuals align with the QQ Line in the central part of the data, but depart substantially from the line at both tails of the data. We can safely conclude that the residuals are not normally distributed, so the assumption is not supported.

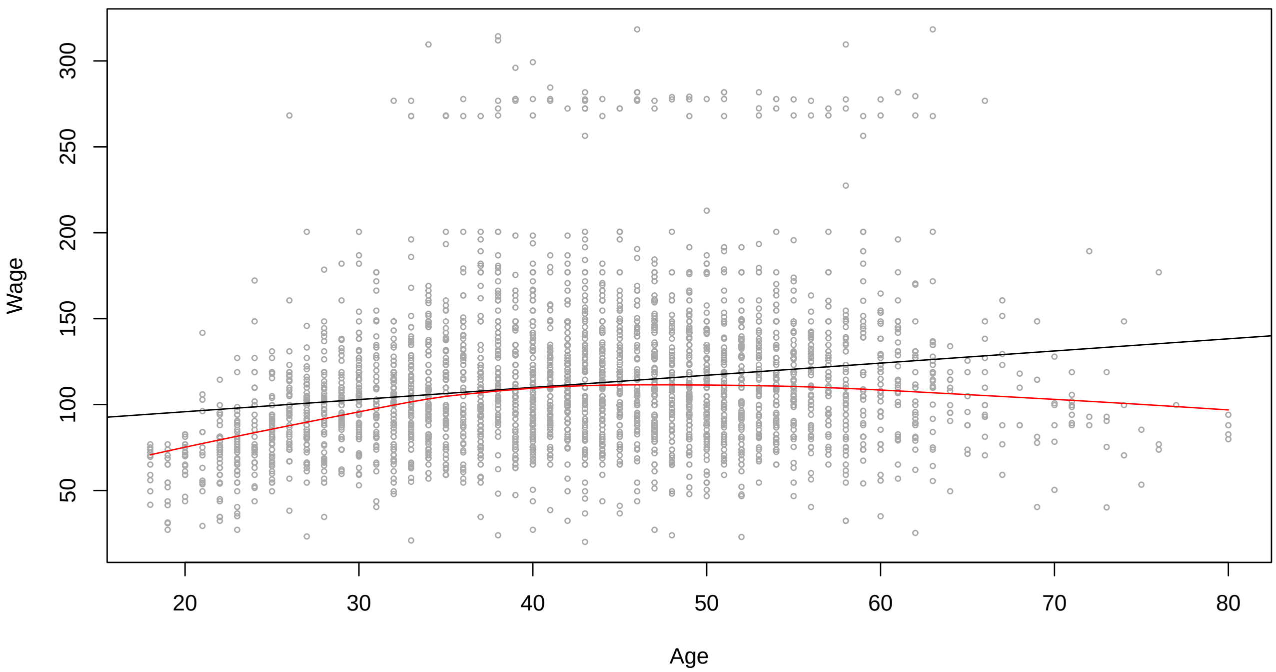
Linearity

Let's use the **Wage** data set in the `{ISLR}` library to illustrate how to inspect for linearity visually.

```
library(ISLR)
lm.fit <- lm(wage ~ age, data = Wage)
```

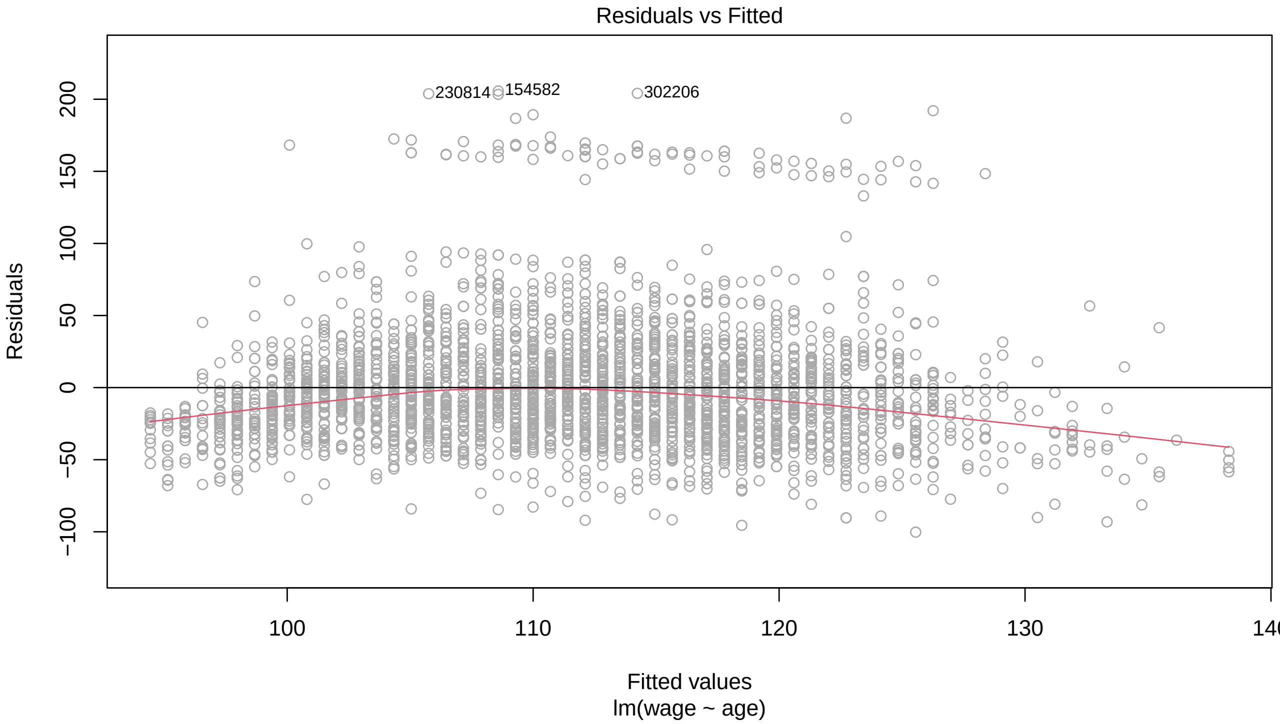
Let's plot wage aginst age and draw through the plot both, a straight line and a fitting curvilinear line. If the relationship between wage and age is linear, both lines should be somewhat overlapping. A substantial departure from the straight line would be a indication of non-linearity:

```
plot(Wage$age, Wage$wage,
     xlab = "Age",
     ylab ="Wage",
     cex=.5,
     col="darkgrey")
abline(lm.fit) # Straight line
lines(lowess(Wage$age, Wage$wage),
      col = "red") # Trend curve
```



Now, let's explore the linearity of the regression model residuals:

```
plot(lm.fit, which = 1,
     col = "darkgrey")
abline(h = 0) # Draw a horizontal line at 0
```

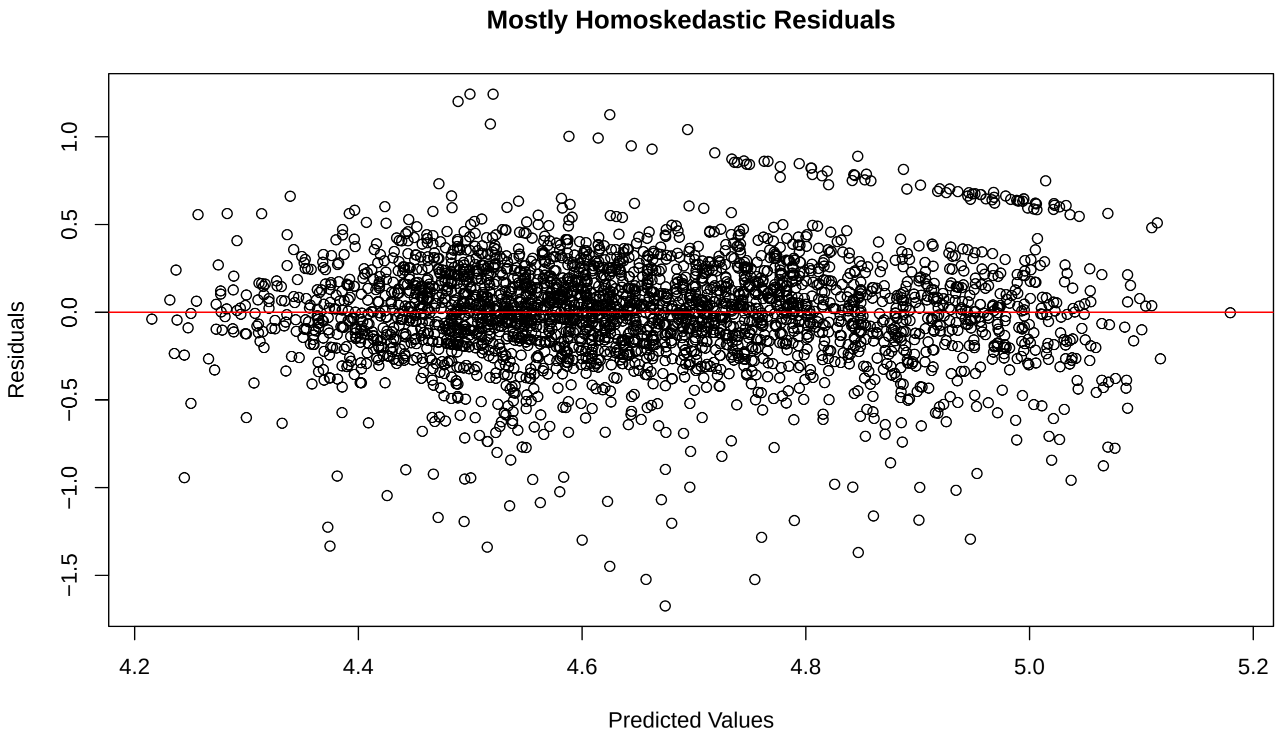


Constant Variance of the Residuals

Residuals are **heteroskedastic** when they vary systematically around the predicted outcome values, thus violating the OLS assumption of constant residual variance. Let's compare homoskedastic with heteroskedastic residuals:

Homoskedastic residuals

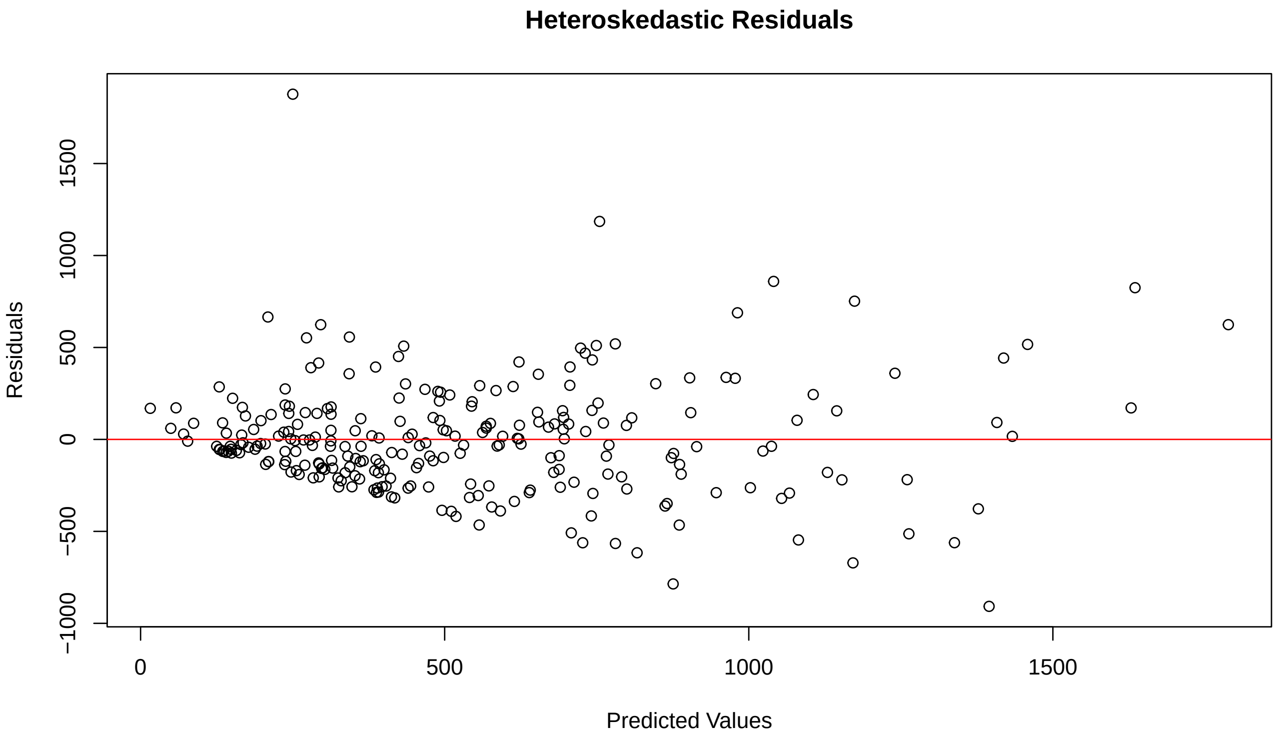
```
library(ISLR) # Contains Wage data set
lm.fit <- lm(logwage ~ year + age + race + education + jobclass,
             data = Wage)
plot(lm.fit$residuals ~ lm.fit$fitted.values,
     main = "Mostly Homoskedastic Residuals",
     xlab = "Predicted Values",
     ylab = "Residuals")
abline(h = 0, col = "red")
```



The plot of residuals against predicted values shows an even distribution of the residuals, so heteroskedasticity is not a concern.

Heteroskedastic Residuals

```
library(ISLR) # Contains the Hitters data set
lm.fit <- lm(Salary ~ ., data = Hitters)
plot(lm.fit$residuals ~ lm.fit$fitted.values,
     main = "Heteroskedastic Residuals",
     xlab = "Predicted Values",
     ylab = "Residuals")
abline(h = 0, col = "red")
```



The plot above shows heteroskedastic residuals, so the OLS assumption of constant residual variance is not supported.

2. Weighted Least Squares (WLS) Regression

Testing for Heteroskedasticity

Look at yourself in the mirror and repeat "heteroskedasticity" about 100 time and you will eventually get the right pronunciation.

The OLS model assumes that residuals are even throughout the regression line. When you fit an OLS model, you need to inspect the first regression plot, which shows residuals against fitted values. The residuals should show a cloud of even data throughout. If it doesn't, then you need to test for **heteroskedasticity** or uneven residuals. If some residuals are much larger than others, then when you square them (to get the sum of error squares), the squared values will be even larger and those observations will pull the regression line disproportionately. When heteroskedasticity is present, we need to weight down the squared residuals to reduce the influence that large residuals (squared) on the regression line.

It is important to note that heteroskedastic residuals don't bias the model because, on average, large positive and negative arrows will compensate each other. However, if we drop a few data points or if we get new data, the model will have too much variance and yield unstable results. To address this issue, we can fit a regression line by minimizing, not the sum of squared residuals, but the **weighted** sum of squared residuals (i.e., **WLS**). The weights we need to use should be inversely proportional to the magnitude of the OLS squared residuals.

There are various weights we can use for WLS, but it is usually best to use the inverse of the OLS errors squared, so that we weight down the sum proportionally to the size of the error squared. That is, the larger the error, the lower the weight we place when computing the regression line that best fits the data. This involves finding a weight vector, which we will call **wts**. This vector has one value for each observation, with the corresponding weight to apply to the residual associated with that observation.

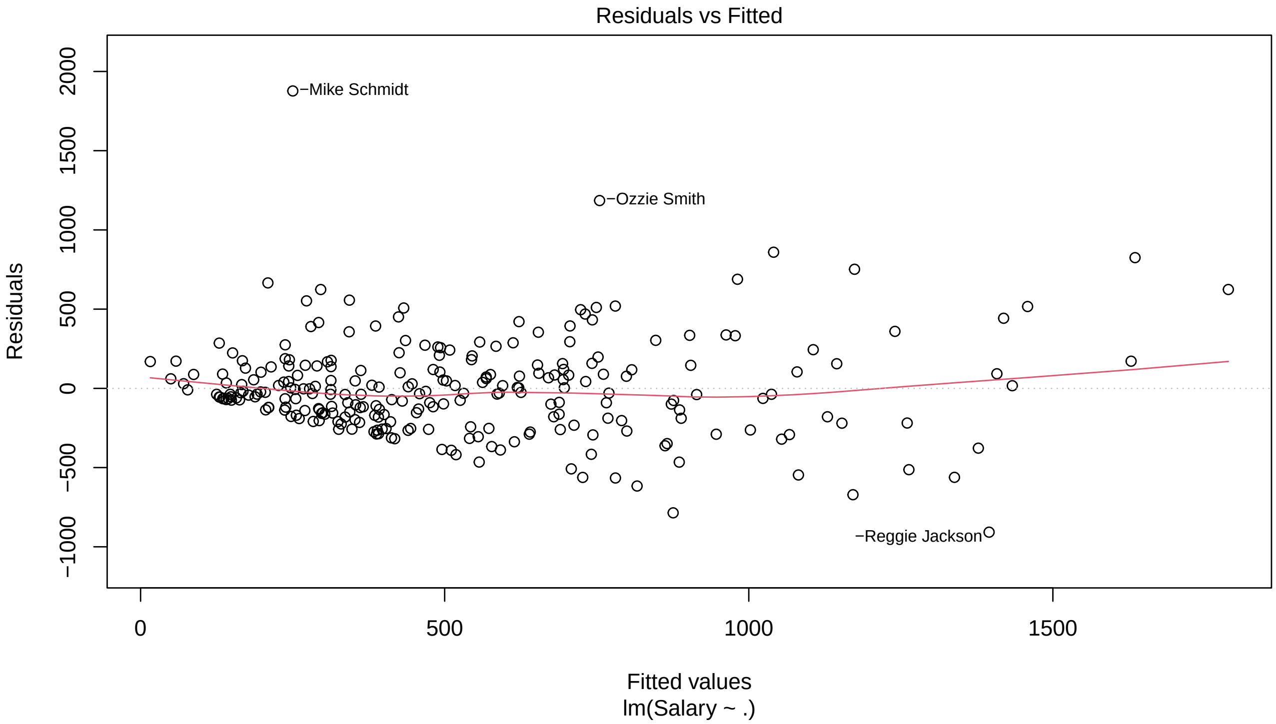
Let's start by fitting an OLS regression model to predict the median value **medv** of homes in the Boston area. There are a few libraries that contain functions to test for heteroskedasticity. We will use **{lmtest}**, which has a suite of tests for `lm()` objects, including a heteroskedasticity test.

```
library(ISLR) # To read the Boston housing market data
library(lmtest) # Contains bptest() and more
options(scipen = 4) # To minimize the use of scientific notation
Hitters <- na.omit(Hitters) # Need to remove several omitted values
lm.ols <- lm(Salary ~ ., data = Hitters)
summary(lm.ols) # Take a look
```

```
Call:
lm(formula = Salary ~ ., data = Hitters)

Residuals:
    Min      1Q  Median      3Q     Max 
-907.62 -178.35  -31.11  139.09 1877.04 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  163.10359   90.77854   1.797 0.073622 .  
AtBat         -1.97987    0.63398  -3.123 0.002008 ** 
Hits           7.50077    2.37753   3.155 0.001808 ** 
HmRun          4.33088    6.20145   0.698 0.485616    
Runs          -2.37621    2.98076  -0.797 0.426122    
RBI           -1.04496    2.60088  -0.402 0.688204    
Walks          6.23129    1.82850   3.408 0.000766 ***
Years         -3.48905   12.41219  -0.281 0.778874    
CAtBat        -0.17134    0.13524  -1.267 0.206380    
CHits          0.13399    0.67455   0.199 0.842713    
CHmRun        -0.17286    1.61724  -0.107 0.914967    
CRuns          1.45430    0.75046   1.938 0.053795 .  
CRBI           0.80771    0.69262   1.166 0.244691    
CWalks        -0.81157    0.32808  -2.474 0.014057 *  
LeagueN       62.59942   79.26140   0.790 0.430424    
DivisionW   -116.84925   40.36695  -2.895 0.004141 ** 
PutOuts        0.28189    0.07744   3.640 0.000333 ***
Assists        0.37107    0.22120   1.678 0.094723 .  
Errors        -3.36076    4.39163  -0.765 0.444857    
NewLeagueN   -24.76233   79.00263  -0.313 0.754218    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 315.6 on 243 degrees of freedom
Multiple R-squared:  0.5461,    Adjusted R-squared:  0.5106 
F-statistic: 15.39 on 19 and 243 DF,  p-value: < 2.2e-16
```

The first step is to inspect the residuals to see if the form a cloud of data around the predicted line, or whether the errors grow or shrink systematically.

```
plot(lm.ols, which = 1)
```



The residuals don't look totally even, but it is not entirely clear that there is heteroskedasticity. To be certain, we will perform a Breusch-Pagan test for Heteroscedasticity using the Breusch-Pagan `bptest(){lmtest}` test. If the p-value is significant this means that the errors squared are highly correlated with the predicted values, which means that the errors increase or decrease systematically. That is, the errors are heteroskedastic.

```
bptest(lm.ols, data = Hitters)
```

```
	studentized Breusch-Pagan test

data:  lm.ols
BP = 32.196, df = 19, p-value = 0.0297
```

The p-value is highly significant, so we reject the null hypothesis of homoskedasticity and conclude the that the residuals are heteroskedastic. Heteroskedasticity does not affect bias of the coefficients. In fact, the resulting coefficients will usually be similar in magnitude to OLS coefficients. But with heteroskedasticity, the OLS model is no longer **BLUE**. That is, it is not the most efficient (i.e., least variance) model and we can use other methods that will result in lower model variance. Therefore, we correct for heteroskedasticity using the **Weighted Least Squares Method (WLS)**.

Fitting a WLS Model

There are many ways to compute the weighted sum of errors squared. A popular and very effective weighting scheme is to use the actual errors of the OLS model to weight down the sum. However, with heteroskedasticity, the residuals may be all over the place, so other weighting schemes may be more appropriate. One other popular approach is to extract the residuals and fitted values from the OLS model and specify a regression model (not a real one, but one just to compute the WLS weights) to predict the absolute value or the squared value of the residuals (this is important because weights in WLS have to be positive values), and then us the inverse value of these fitted values as weights. So, here are some options:

Known Predictor: If you have knowledge of a predictor X that is causing the heteroskedasticity, you can use the respective X values or 1/X values as weights, depending on whether X is causing errors to shrink (use X) or to grow (use 1/X). The idea is to weight down large errors.

Fitted Values of OLS Residuals: We often don't know which predictors may be causing heteroskedasticity. But using OLS residuals is an effective methods when the errors grow or shrink systematically along the fitted values and when we don't know the X causing the problems. Since residuals can be positive or negative, it is necessary to either use the absolute value of the residuals, i.e., `1 / abs(fitted(residuals))`, or the squared values of the residuals, i.e., `1 / residuals ^ 2`. We take the inverse so that large residuals have smaller weight on the regression estimation than smaller residuals. There are two ways to do this. I Illustrate one way using the squared residuals and the other way using absolute values, in steps:

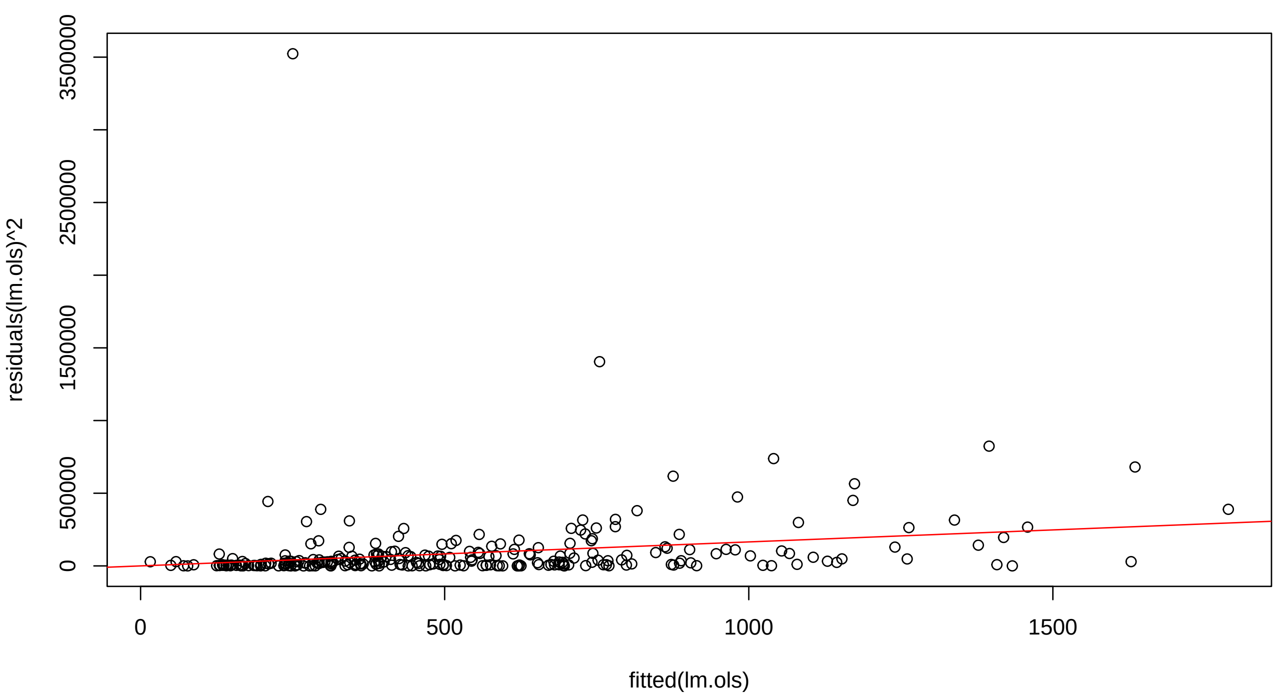
Step 1: Fit an OLS model

We did this above already, yielding the **lm.ols** object.

Step 2:

Fit another linear model to predict the squared errors with fitted (i.e., predicted) values of the OLS model above as predictors. I also provide a visualization for your reference. Notice that I added **0** as a **predictor**. This forces the model to have no intercept, so the lm.res2 regression line starts at the origin. This is not necessary in many cases, but in a few cases, some fitted values of lm.res2 at the far left of the fitted line may be negative. Forcing the regression to start at the origin resolves this issue.

```
lm.res2 <- lm(residuals(lm.ols) ^ 2 ~ 0 + fitted(lm.ols))
# If you prefer to usde absolute values, you can fit this model instead:
# lm.res.abs <- lm(abs(residuals(lm.ols)) ~ 0 + fitted(lm.ols))

plot(fitted(lm.ols), residuals(lm.ols) ^ 2) # Take a look
abline(lm.res2, col = "red") # Draw regression line
```



We will be using the predicted values values along the red regression line of the predicted squared errors as weights the, rather than the actual errors squared. We could use the actual error squares, but notice the wide variance of the errors in the plot above. You could still fit a WLS model with the actual error squares, but research has shown that using the predicted values instead yield more stable results. Naturally, we want to use the predicted error squares to **weight down** their influence in the SSE calculation, so we will use their inverse for that purpose:

```
wts <- 1 / fitted(lm.res2)
# or wts <- 1 / fitted(lm.res.abs) # If you use absolute values
```

Also, if you did not use the 0 as a predictor as indicated in the previous code chunk, you can avoid the problem with negative weights by simply making the **wts** values positve this way:

wts <- abs(wts)

```
wts <- abs(wts)
```

Actual Residuals: A simpler approach that works well when residuals in the OLS model grow or shrink monotonically is to use the inverse of the actual residuals as weights, either squared or in absolute values. Both methods are effective at correcting for heteroskedasticity. As long as you weight down large residuals, you will get a better model. For example:

```
wts <- 1 / residuals(lm.ols) ^ 2 # or
wts <- 1 / abs(residuals(lm.ols)) # or
```

Once you have computed the **wts** weight vector, use it as a parameter in the WLS model:

```
lm.wls <- lm(Salary ~ .,
             data = Hitters,
             weights = wts)
summary(lm.wls)
```

```
Call:
lm(formula = Salary ~ ., data = Hitters, weights = wts)

Weighted Residuals:
    Min      1Q  Median      3Q     Max
-2.0266 -0.6785 -0.1944  0.4599  9.0591

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 144.62645   68.46812   2.112  0.03568 *
AtBat        -1.23615    0.50429  -2.451  0.01494 *
Hits          4.64320    2.33718   1.987  0.04808 *
HmRun         7.52922    5.18572   1.452  0.14782
Runs         -1.31828    2.72036  -0.485  0.62840
RBI          -1.89272    2.10699  -0.898  0.36991
Walks         3.72630    1.66159   2.243  0.02582 *
Years        -0.10555   10.21262  -0.010  0.99176
CAtBat       -0.14002    0.14400  -0.972  0.33186
CHits         0.50544    0.74819   0.676  0.49997
CHmRun        1.04653    1.81865   0.575  0.56553
CRuns         0.97578    0.78210   1.248  0.21336
CRBI         -0.02541    0.77619  -0.033  0.97391
CWalks       -0.64497    0.35652  -1.809  0.07167 .
LeagueN      81.55654   66.43973   1.228  0.22081
DivisionW   -78.27314   34.26967  -2.284  0.02323 *
```

```
PutOuts       0.26695    0.08292   3.219  0.00146 **
Assists       0.52664    0.20200   2.607  0.00970 **
Errors       -5.66111    3.79055  -1.493  0.13661
NewLeagueN  -26.67731   66.96025  -0.398  0.69068
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.088 on 243 degrees of freedom
Multiple R-squared:  0.496, Adjusted R-squared:  0.4566
F-statistic: 12.59 on 19 and 243 DF,  p-value: < 2.2e-16
```

Compared to the OLS regression model, the WLS model has a slightly lower R-squared, but it is still very high and significant. The WLS model generally yields a higher R-squared, but not always as in this case. Also notice that the coefficients have not changed much in terms of magnitude, mainly because both models are unbiased, so they should yield similar predictor effects. But the significance of the predictors has changed more. Generally, the WLS model will show smaller p-values, that is, more significant effects, although not always, as you will be able to see in the output. But the main thing is that the WLS model is more stable than the OLS model. That is, it has less variance. So, if we fit a model with multiple sub-samples (as we do with machine learning), the WLS model will yield more consistent results than the OLS model.

However, if the resulting WLS model still suffers from heteroskedasticity, you could do another iteration by using the errors of the WLS model to fit the subsequent WLS model, and so on. This would get pretty tedious. But approach is actually called **Iterative Re-Weighted Least Squares (IRLS)**. The WLS method described above is pretty standard. But there are other methods and various weighting schemes you can use. Iterative Re-Weighted Least Squares (IRLS) is another popular method to compute **robust** estimators when heteroscedasticity is present. The **{MASS}** library has a **Residual Linear Model** function `rlm()`, which does OLS with robust residuals. This method is similar to WLS, with one difference, which is, because the weights are calculated from residuals, but the final residuals are dependent on these weights, the IRLS solves this issue by itereating the model several times, calculating the residuals each subsequent time, re-weighting the model, and so on – i.e., running WLS multiple times in iterations until the residuals don't change any more (the model converges). Most of the time WLS will be sufficient. But if WLS does not solve the heteroskedasticity problem, IRLS will continue iterating further weighted models until heteroskedasticity is resolved. You can fit an IRLS model this way:

```
rlm.fit <- rlm(Salary ~ ., data = Hitters)
summary(rlm.fit) # Take a look
```

3. Generalized Linear Method (GLM)

Overview

The `lm()` function fits a linear model using OLS. It assumes that the **residuals are normally distributed**. The outcome variable does not have to be normally distributed, as long as the residuals are normally distributed. However, when the outcome variable is not normally distributed, the residuals are almost never normally distributed. Since you can inspect the normality of the outcome variable before you run your regression model, this is generally one of the first things to do.

If the outcome variable is not normally distributed, but you can ascertain the type of distribution it follows (e.g., poisson, logistic), then you can use the `glm()` function instead of the `lm()` to fit your model. But the `glm()` model requires that you specify the distribution family of the residuals.

The `glm()` function works just like the linear models function `lm()`, except that it can be used to fit models for a wider range of distributions, not just normal. It does not use the formulas to minimize the SSE, but it uses the **Maximum Likelihood Estimation (MLE)** method. Interesting, it can be shown mathematically that if you fit a `glm()` model using a **Gaussian** family distribution (omitted below because it is the default), you get the exact same results, except the `lm()` reports OLS fit statistics (e.g. R-Squared, ANOVA, etc.), whereas `glm()` reports **MLE** fit statistics (e.g., 2LL, Deviance, etc.). Take a look:

OLS Model

```r
library(MASS) # Needed for the Boston data set
lm.fit <- lm(medv ~ lstat + age, data = Boston) # OLS
summary(lm.fit) # Check it out
```

```

Call:
lm(formula = medv ~ lstat + age, data = Boston)

Residuals:
    Min      1Q  Median      3Q     Max 
-15.981  -3.978  -1.283   1.968  23.158 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 33.22276    0.73085  45.458  < 2e-16 ***
lstat       -1.03207    0.04819 -21.416  < 2e-16 ***
age          0.03454    0.01223   2.826  0.00491 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.173 on 503 degrees of freedom
Multiple R-squared:  0.5513,    Adjusted R-squared:  0.5495 
F-statistic:   309 on 2 and 503 DF,  p-value: < 2.2e-16
```

GLM Model

```r
glm.fit = glm(medv ~ lstat + age,
              data = Boston) # GLM
summary(glm.fit)
```

```

Call:
glm(formula = medv ~ lstat + age, data = Boston)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 33.22276    0.73085  45.458  < 2e-16 ***
lstat       -1.03207    0.04819 -21.416  < 2e-16 ***
age          0.03454    0.01223   2.826  0.00491 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 38.10761)

    Null deviance: 42716  on 505  degrees of freedom
Residual deviance: 19168  on 503  degrees of freedom
AIC: 3283

Number of Fisher Scoring iterations: 2
```

Notice that You can derive the R-square of a GLM model by computing how much the deviance is improving as a percentage of the null model's deviance:

```r
res.dev <- glm.fit$deviance
null.dev <- glm.fit$null.deviance

# Proportion of deviance explained by the model:
dev.Rsq <- (null.dev - res.dev) / null.dev
print(dev.Rsq, digits = 4)
```

```
[1] 0.5513
```

Binomial Logistic Regression

Binomial Logit models are used to predict binary outcomes (e.g., yes/no, accept/reject, approve/decline). For example, let's read the South Africa heart disease data set from the authors' data site:

```r
heart <- read.table("Heart.csv",
                    sep = ",",
                    head = T)
```

The dataset documentation is available at: https://web.stanford.edu/~hastie/ElemStatLearn/datasets/SAheart.info.txt

In the example above, I used a Heart.csv I had already downloaded from the ISLR website, but you could read the file directly from the web with this command (commented out for now):

```r
heart <- read.table(
   "https://web.stanford.edu/~hastie/ElemStatLearn/datasets/SAheart.data",
                    sep = ",",
                    head = T,
                    row.names = 1)
```

You can view other datasets in the ISLR book's website: https://web.stanford.edu/~hastie/ElemStatLearn/datasets/

Now let's fit a binomial Logistic regression model:

```r
heart.fit <- glm(chd ~ .,
                 data = heart,
                 family = binomial(link = "logit"))
```

chd is 1 if patient has coronary heart disease, 0 if not

The family **distribution** of the outcome variable is `family = binomial`

The **link** function is what we use when we want to transform the outcome variable. When we fitted the OLS model with `glm()` above we used `link = "identity"`, which is the default and it means **do nothing**, so we omitted it. But to fit the binomial Logistic model we need to transform the outcome from binary to log-likelihoods using the Logistic function or `link = "logit"`. Let's take a look at the results:

```r
summary(heart.fit)
```

```

Call:
glm(formula = chd ~ ., family = binomial(link = "logit"), data = heart)

Coefficients:
                 Estimate Std. Error z value   Pr(>|z|)
```

```
(Intercept)    -6.1507209  1.3082600  -4.701 0.00000258 ***
sbp             0.0065040  0.0057304   1.135   0.256374    
tobacco         0.0793764  0.0266028   2.984   0.002847 ** 
ldl             0.1739239  0.0596617   2.915   0.003555 ** 
adiposity       0.0185866  0.0292894   0.635   0.525700    
famhistPresent  0.9253704  0.2278940   4.061 0.00004896 ***
typea           0.0395950  0.0123202   3.214   0.001310 ** 
obesity        -0.0629099  0.0442477  -1.422   0.155095    
alcohol         0.0001217  0.0044832   0.027   0.978350    
age             0.0452253  0.0121298   3.728   0.000193 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 596.11  on 461  degrees of freedom
Residual deviance: 472.14  on 452  degrees of freedom
AIC: 492.14

Number of Fisher Scoring iterations: 5
```

Interpretation

The coefficients represent the **change** in log-odds (i.e., logit) that the outcome variable is `= 1`. A positive coefficients indicates how much the log-odds (and the odds) of Y being 1 increase, when the variable goes up by 1. A negative coefficient indicates how much the log-odds decrease.

Again, it is always useful to convert the log-odds coefficients to odds, because log-odds are very hard to explain to a manager or business client. Let's convert the log-odds coefficients to odds and insert them in the summary output.

Let's start by creating an object with the contents of `summary(heart.fit)`:

```
lm.sum <- summary(heart.fit)
```

Now let's extract and display some fit statistics.

** Null Model Deviance

```
null.dev <- paste("Null Deviance",
                  round(lm.sum$null.deviance, digits = 2),
                  "on", lm.sum$df.null, "degrees of freedom")
```

** Fitted Model Residual Deviance

```
res.dev <- paste("Residual Deviance",
                 round(lm.sum$deviance, digits = 2),
                 "on", lm.sum$df.residual, "degrees of freedom")
```

AIC Fit Statistic

```
aic <- paste("AIC", round(lm.sum$aic, digits = 2))
```

Display All Fit Statistics Together

```
cat(null.dev, res.dev, aic, sep = "\n")
```

```
Null Deviance 596.11 on 461 degrees of freedom
Residual Deviance 472.14 on 452 degrees of freedom
AIC 492.14
```

We can compute the proportion of reduced deviance by the model from the null model (similar to R-Squared) as (Null Deviance - Residual Deviance) / (Null Deviance):

```
red.dev <- (lm.sum$null.deviance - lm.sum$deviance) / lm.sum$null.deviance
cat("Proportion of reduced deviance:", round(red.dev, digits = 4))
```

```
Proportion of reduced deviance: 0.208
```

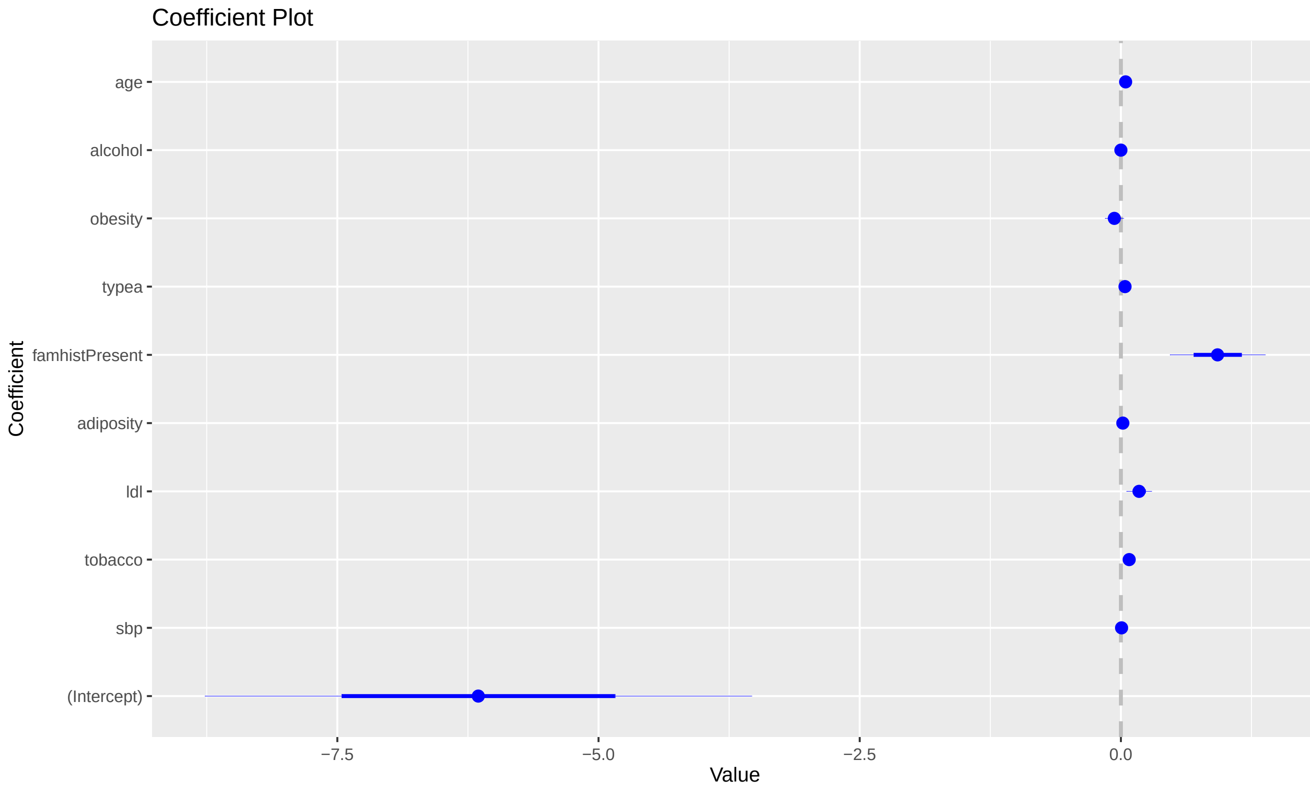
Log-Odds and Odds Coefficients

```
round(
  cbind("Log Odds" = lm.sum$coefficients[ ,1],
        "Odds" = exp(lm.sum$coefficients[ ,1]),
         lm.sum$coefficients[ ,2:4]),
         digits = 4)
```

```
               Log Odds   Odds Std. Error z value Pr(>|z|)
(Intercept)     -6.1507 0.0021     1.3083 -4.7015   0.0000
sbp              0.0065 1.0065     0.0057  1.1350   0.2564
tobacco          0.0794 1.0826     0.0266  2.9838   0.0028
ldl              0.1739 1.1900     0.0597  2.9152   0.0036
adiposity        0.0186 1.0188     0.0293  0.6346   0.5257
famhistPresent   0.9254 2.5228     0.2279  4.0605   0.0000
typea            0.0396 1.0404     0.0123  3.2138   0.0013
obesity         -0.0629 0.9390     0.0442 -1.4218   0.1551
alcohol          0.0001 1.0001     0.0045  0.0271   0.9784
age              0.0452 1.0463     0.0121  3.7285   0.0002
```

You can inspect the coefficients visually too:

```
require(coefplot)
coefplot(heart.fit)
```



I will interpret a positive and a negative coefficient, as an example.

Effect of **tobacco**. This variable measures the cumulative consumption of tobacco in kilograms. Holding everything else constant, on average, an increase in 1 Kg. of cumulative consumption of tobacco increases the **log-odds** of developing coronary heart disease by 0.0794 and this effect is significant. Also, the **odds** of developing coronary heart disease increases by a **factor of 1.082**. Notice the use of the term **factor of**. This is a very important aspect of the interpretation. Log-odds are additive because they are in the linear model. So the log-odds effect reflect unit changes, just like any OLS coefficient. But when we convert log-odds to odds, we are using the `exp()` function, so the odds are now multiplicative, so their effect is a multiplication of the odds. So, odds > 1 than 1 increase the odds (just like positive log-odds) and odds < 1 reduce the odds (just like negative log-odds)

4. Decision Trees

Decision trees are non-parametric models that simply split the data at points called **knots** into **regions** or tree **leaves**, based on a predictor and the corresponding value that does the best job at reducing the deviance in the outcome variable predictions, and continue to split the leaves into further sub-regions recursively. A **regression tree** is the non-parametric equivalent to a linear regression model in which the prediction is a quantitative value. A **classification tree** is the non-parametric equivalent to a Logistic regression, and other classification methods, in which the prediction is the correct classification of the outcome into a category. When splitting a tree region into further sub-regions, a regression tree aims to reduce the mean squared error within regions, whereas a classification tree aims to reduce the mis-classification of outcomes into categories. In either case, the tree splits the data recursively based on predictors and threshold values that reduce deviance the most.

Regression Trees

Say, for example, that you have 2 predictors. A regression tree will find which of the two predictors does a better job at separating the outcome values into two regions, and finds the value of that predictor in which the mean squared error MSE within each region is minimized. The regression tree model assigns a predicted value for each region equal to the mean of the outcome value withing that region. It then continues recursively, taking each region and further splitting them into more regions or leaves.

The criteria for evaluating which variable to use for the split and at which value is based on minimizing the MSE. The process continues with further subdivisions of the data. Each sub-division portion is called a “leaf” and the splitting point is called a “node”. In principle, one could continue the data splitting until each data point is a leaf, but this would over fit the data.

The next example fits a regression tree to predict median values of houses in Boston suburbs.We first need to load the **{tree}** library.

```
library(MASS) # Contains the Boston data set
library(tree) # Needed to fit decision trees
tree.boston <- tree(medv ~ ., Boston)
```

Let’s view the summary results (the **residual mean deviance** is the **total residual deviance** (i.e., the sum of squares of the residuals) divided by the total number of observations):

```
summary(tree.boston)
```

```
Regression tree:
tree(formula = medv ~ ., data = Boston)
Variables actually used in tree construction:
[1] "rm"      "lstat"   "dis"     "crim"    "ptratio"
Number of terminal nodes:  9
Residual mean deviance:  13.55 = 6734 / 497
Distribution of residuals:
     Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
-17.68000  -2.23000   0.07026   0.00000   2.22100  16.50000
```

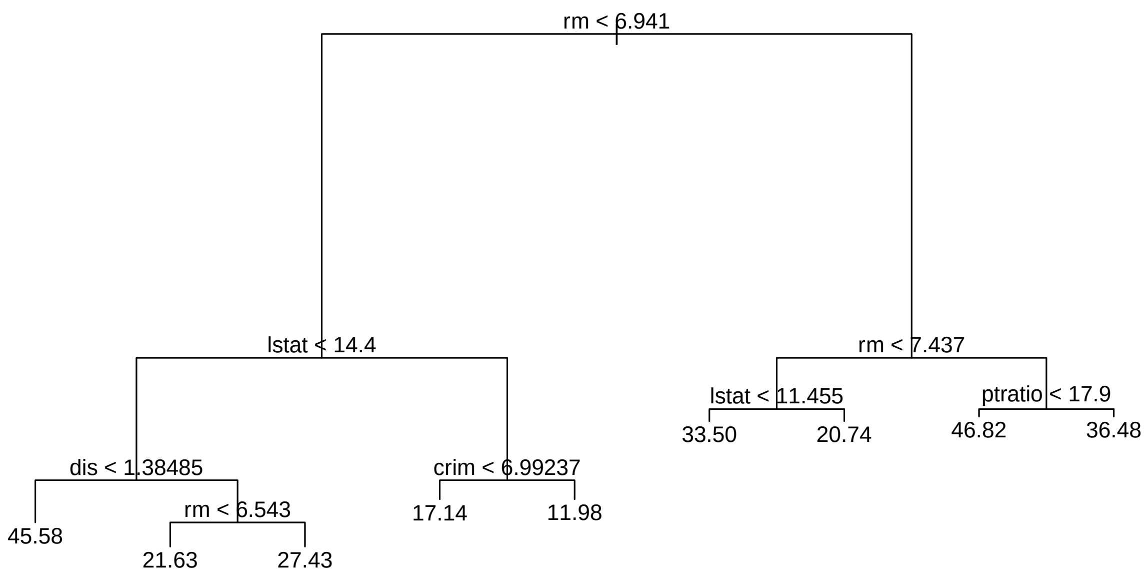
Let’s view the tree nodes and branches:

```
tree.boston
```

```
node), split, n, deviance, yval
      * denotes terminal node

 1) root 506 42720.0 22.53
   2) rm < 6.941 430 17320.0 19.93
     4) lstat < 14.4 255  6632.0 23.35
       8) dis < 1.38485 5   390.7 45.58 *
       9) dis > 1.38485 250  3721.0 22.91
        18) rm < 6.543 195  1636.0 21.63 *
        19) rm > 6.543 55   643.2 27.43 *
     5) lstat > 14.4 175  3373.0 14.96
      10) crim < 6.99237 101  1151.0 17.14 *
      11) crim > 6.99237 74  1086.0 11.98 *
   3) rm > 6.941 76  6059.0 37.24
     6) rm < 7.437 46  1900.0 32.11
      12) lstat < 11.455 41   844.2 33.50 *
      13) lstat > 11.455 5   329.8 20.74 *
     7) rm > 7.437 30  1099.0 45.10
      14) ptratio < 17.9 25   340.7 46.82 *
      15) ptratio > 17.9 5   312.7 36.48 *
```

Now lets visualize the tree:

```r
plot(tree.boston) # Plot the tree
text(tree.boston, pretty = 0) # Let's make it pretty and add labels
```



Controlling the Tree Size

Trees can grow quite large with large data sets. It is very important to figure out the optimal tree size to **grow** (forward) or **prune** (backward). I will discuss later how to find the optimal size of a tree using cross-validation MSE testing, which will yield the tree with highest predictive accuracy. But, for now, let's discuss how to control the size of the tree using the `mindev =` attribute.

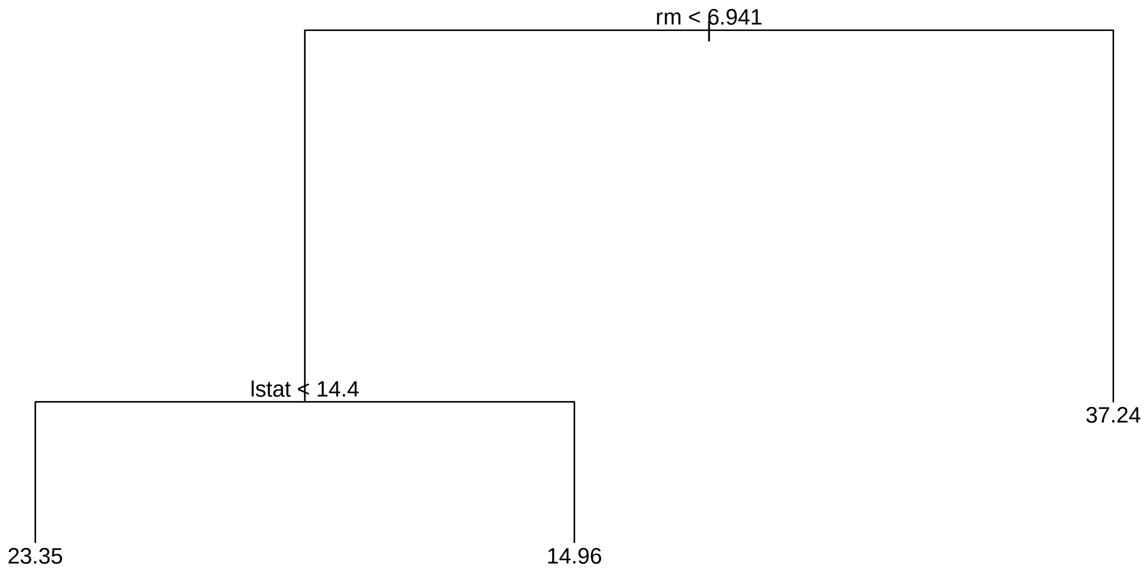
Before you start splitting the tree you have the root of the tree, which is equivalent to the **null** model with no predictors. At the root, the prediction for all data points is equal to the mean of the outcome variable for all the data points. Of course, we can do much better than this. After the first split, we have 2 leaves. We then predict all observations within a leaf to be equal to the mean of the outcome variable for all data points within that leaf. The MSE is calculated by squaring the differences between the actual outcome values and the outcome mean within the corresponding leaf.

As you would expect, the MSE gets reduced every time we split the tree into further regions. But if we keep splitting all the way till the end, we will end up with **terminal nodes**, in which we will have one leaf for each data point. The tree will be perfectly accurate because the predicted value will be identical to the actual value. Naturally, this accuracy is artificial because the tree is over-fitting and will not necessarily predict well when new data arrives. Cross-validation testing can help us find the optimal tree size in which predictions are most accurate with new data.

For now, we will simply use a parameter called `mindev =` to control the size of the tree. By default the `tree()` function stops growing the tree at `mindev = 0.01`. That is, when the MSE of the tree is 0.01 (or 1%) of the MSE at the root, the tree stops growing. We can **reduce** the `mindev` value to let the tree grow **larger** (i.e., it will take longer to reduce the MSE to that level) or **increase** it to grow the tree smaller (i.e., it will take less time to reduce the MSE to that level).

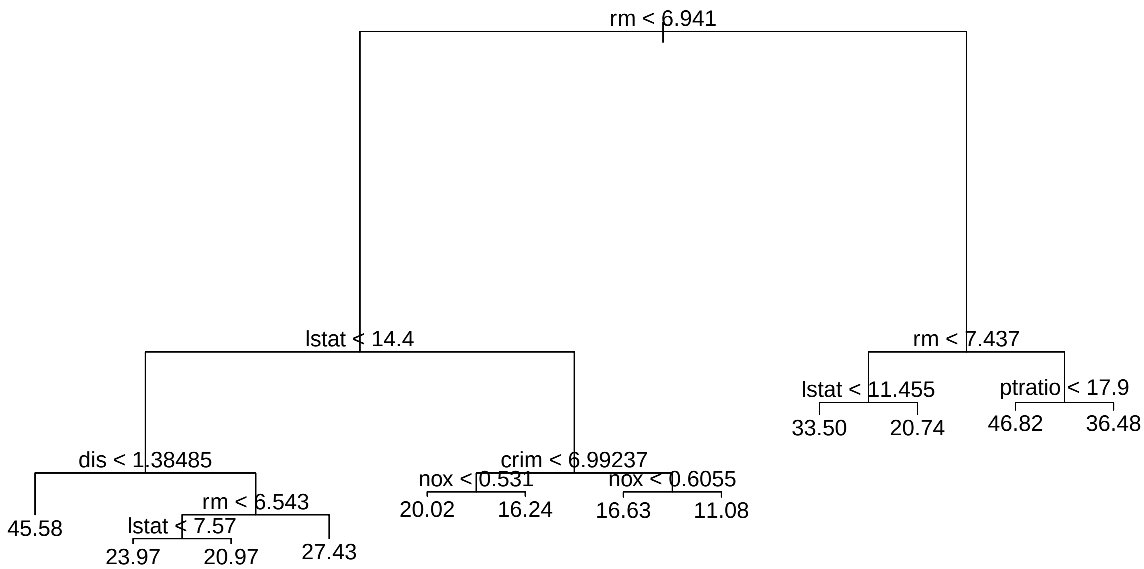
Small Tree Example, `mindev > 0.01`

```r
tree.boston.small <- tree(medv ~ ., Boston,
                          mindev = 0.1)
plot(tree.boston.small)
text(tree.boston.small, pretty = 0)
```



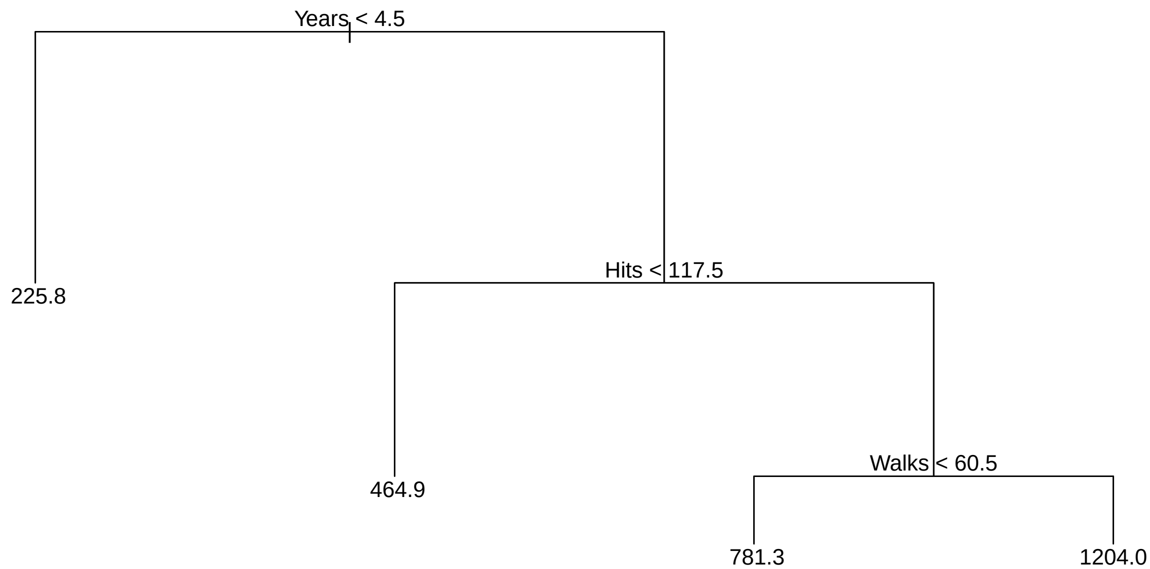
Large Tree Example, `mindev < 0.01`

```r
tree.boston.large <- tree(medv ~ ., Boston,
                          mindev = 0.005)
plot(tree.boston.large)
text(tree.boston.large, pretty = 0)
```



Example with Baseball Player Salary Data

```r
library(ISLR)
tree.sal <- tree(Salary ~
                 Years + Hits + RBI + PutOuts + Runs + Walks,
                 Hitters,
                 mindev = 0.05)
plot(tree.sal)
text(tree.sal, pretty = 0)
```



Classification Trees (Binomial)

Classification trees work just like regression trees, but the response variable is binary (i.e., a classification). While decision are generally not as precise as logistic regression models, and despite the fact that they are not very useful for interpretation there is an abundance of sophisticated decision tree methods (e.g., Bootstrap Aggregation, Random Forests, Boosting, etc.), which can be quite accurate for prediction.

Let's illustrate a classification tree with the **Carseats** data set in the **{ISLR}** package. The data set contains simulated data of child car seat sales in 400 stores. Since the outcome variable **Sales** is quantitative, let's create a binary named **HighSales** with a value of **No** if `Sales < 8000` units and **Yes** otherwise (note that Sales is in thousands of units) and save it in a new data frame named **Carseats.hi**. We then add the HighSales variable vector to the Carseats data frame.

```
library(ISLR)
library(tree)
HighSales <- as.factor(ifelse(Carseats$Sales <= 8,
                              "No", "Yes"))
Carseats.hi <- data.frame(Carseats, HighSales)
```

We can now fit the classification tree. But note that we need to exclude the variable **Sales** from the model because otherwise it would be redundant with (i.e., mathematically related to) **HighSales**. Also, I set `mindev =` arbitrarily to get a good visual on the tree graph.

```
tree.carseats <- tree(HighSales ~ . -Sales,
                      Carseats.hi,
                      mindev = 0.02)
summary(tree.carseats)
```

```
Classification tree:
```

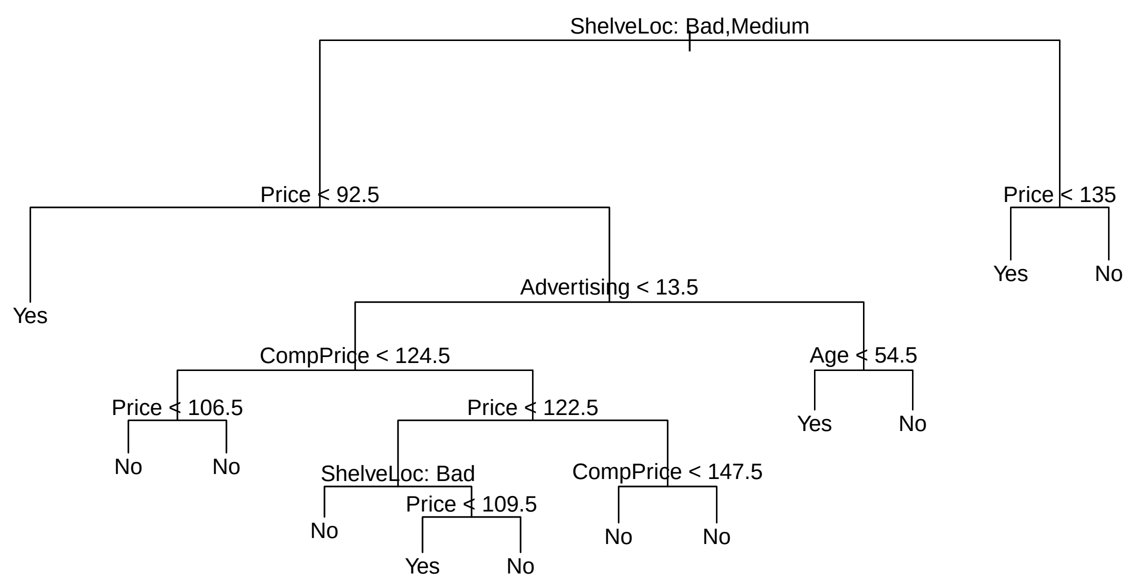
```
tree(formula = HighSales ~ . - Sales, data = Carseats.hi, mindev = 0.02)
Variables actually used in tree construction:
[1] "ShelveLoc"   "Price"       "Advertising" "CompPrice"   "Age"
Number of terminal nodes:  12
Residual mean deviance:  0.7673 = 297.7 / 388
Misclassification error rate: 0.1625 = 65 / 400
```

Notes:

- The **Number of Terminal Nodes** is the number of branches at the bottom end of the tree due to either the resulting tree size from the mindev parameter, or the tree size when all branches are pure (i.e., can no longer continue splitting the tree).
- The **residual mean deviance** in classification trees is the **total residual deviance** (i.e., the sum of squares of the residuals) divided by the (total number of observations - number of terminal nodes). This statistic is not as useful with classification trees (as it is for regression trees), as the sum of squares of the residual is not as meaningful. However, they it is useful to compare various tree models -> the model with the smallest deviance (between actual and predicted values is better). Having said that, it is much better to compare trees using cross-validation, as explained in Chapter 10.
- The **mis-classification error** is more meaningful. It is the total number of **mis-classified observations**, relative to the **total** number of **observations**.

Now let's analyze the tree visually

```
plot(tree.carseats) # Display tree
text(tree.carseats, pretty = 0) # Display data labels and make it pretty
```




```
tree.carseats # Display the data for every leaf
```

```
node), split, n, deviance, yval, (yprob)
      * denotes terminal node

  1) root 400 541.500 No ( 0.59000 0.41000 )
    2) ShelveLoc: Bad,Medium 315 390.600 No ( 0.68889 0.31111 )
      4) Price < 92.5 46  56.530 Yes ( 0.30435 0.69565 ) *
      5) Price > 92.5 269 299.800 No ( 0.75465 0.24535 )
       10) Advertising < 13.5 224 213.200 No ( 0.81696 0.18304 )
         20) CompPrice < 124.5 96  44.890 No ( 0.93750 0.06250 )
           40) Price < 106.5 38  33.150 No ( 0.84211 0.15789 ) *
           41) Price > 106.5 58   0.000 No ( 1.00000 0.00000 ) *
         21) CompPrice > 124.5 128 150.200 No ( 0.72656 0.27344 )
           42) Price < 122.5 51  70.680 Yes ( 0.49020 0.50980 )
             84) ShelveLoc: Bad 11   6.702 No ( 0.90909 0.09091 ) *
             85) ShelveLoc: Medium 40  52.930 Yes ( 0.37500 0.62500 )
              170) Price < 109.5 16   7.481 Yes ( 0.06250 0.93750 ) *
              171) Price > 109.5 24  32.600 No ( 0.58333 0.41667 ) *
           43) Price > 122.5 77  55.540 No ( 0.88312 0.11688 )
             86) CompPrice < 147.5 58  17.400 No ( 0.96552 0.03448 ) *
             87) CompPrice > 147.5 19  25.010 No ( 0.63158 0.36842 ) *
       11) Advertising > 13.5 45  61.830 Yes ( 0.44444 0.55556 )
         22) Age < 54.5 25  25.020 Yes ( 0.20000 0.80000 ) *
         23) Age > 54.5 20  22.490 No ( 0.75000 0.25000 ) *
    3) ShelveLoc: Good 85  90.330 Yes ( 0.22353 0.77647 )
      6) Price < 135 68  49.260 Yes ( 0.11765 0.88235 ) *
      7) Price > 135 17  22.070 No ( 0.64706 0.35294 ) *
```